

* Coordination Compounds *



→ Transition metals have tendency to form large no. of complex compound (coordination compd)

Central metal atom Ligand

→ comp. in which metal atom is bonded to anion/neutral molecule by coordinate bond is coordinate comp.

Chlorophyll - Mg

Haemoglobin - Fe

Vitamin B₁₂ - Co

* Werner's Theory

First scientist to study about the structure

Postulates of Werner theory

i) Metals have 2 types of valency

i) Primary valency

→ (Oxidation no., ionizable, satisfied by -ve ligand. non-directional)

→ $\text{CrCl}_3 \rightarrow 3$

$\text{CoCl}_2 \rightarrow 2$

$\text{PdCl}_2 \rightarrow 2$

ii) Secondary valency

→ Coordination no., non-ionizable, satisfied by -ve charge, neutral directional, gives definite geometry

→ $\text{CoCl}_3 \cdot 6\text{NH}_3$ $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3 \rightarrow 3 \text{ mol AgCl}$ Yellow

$\text{CoCl}_3 \cdot 5\text{NH}_3$ $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 \rightarrow 2 \text{ mol AgCl}$ Purple

$\text{CoCl}_3 \cdot 4\text{NH}_3$ $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \rightarrow 1 \text{ mol AgCl}$ Green

$\text{CoCl}_3 \cdot 4\text{NH}_3$ $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \rightarrow 1 \text{ mol AgCl}$ Violet

coordination polyhedra - ion / Group bonded by secondary linkage with central atom have different spatial arrangement called coordination polyhedra

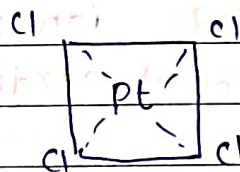
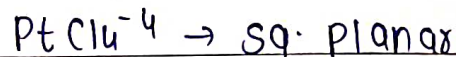
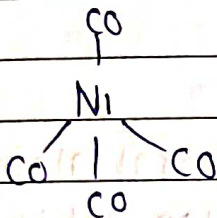
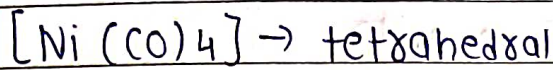
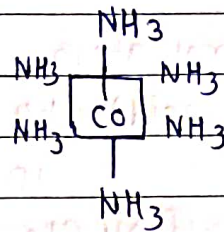
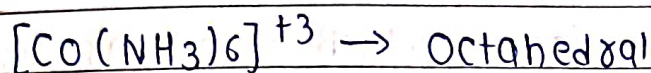


tetrahedral



Octahedral

$\begin{bmatrix} & & \\ & \text{---} & \\ & & \end{bmatrix}$ Sq. planar

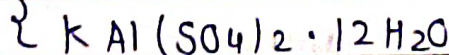
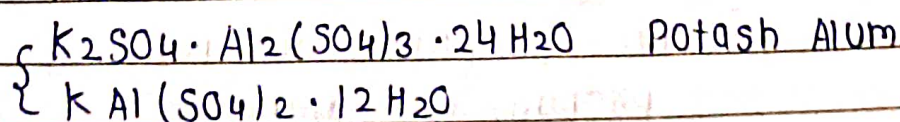
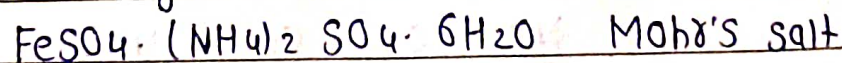
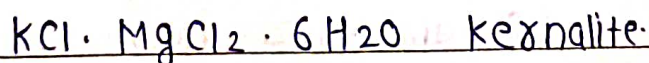


coordination entity $\rightarrow$ Species inside sq. bracket

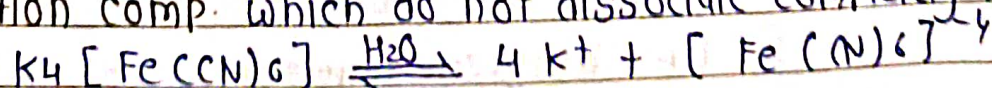
counter ion $\rightarrow$ outside the sq. bracket

Difference betⁿ Double salt & coordination comp

Double salt - when dissolved in water - completely breaks down into ions



coordination comp. which do not dissociate completely into H_2O



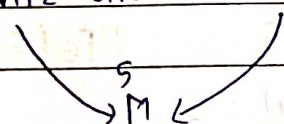


-4 charge

$$\text{NH}_2 - \overset{\text{O}}{\text{C}}\text{H}_2 - \text{CH}_2 - \text{NH} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$$

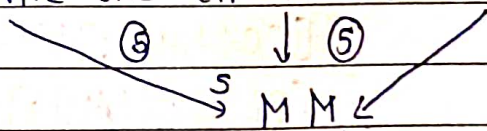
Tridentate

tyjen: $\text{NH}_2-\text{CH}_2-\text{CH}_2-\text{NH}-\text{CH}_2-\text{CH}_2-\text{NH}-\text{CH}_2-\text{CH}_2-\text{NH}_2$

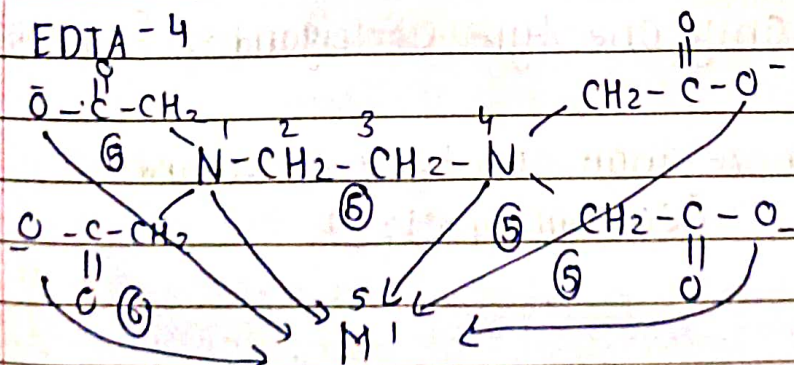
$$\text{en} \rightarrow \overset{1}{\text{NH}_2} - \overset{2}{\text{CH}_2} - \overset{3}{\text{CH}_2} - \overset{4}{\text{NH}_2}$$


5 member ring

dien- $\overset{1}{\text{NH}_2} - \overset{2}{\text{CH}_2} - \overset{3}{\text{CH}_2} - \overset{4}{\text{NH}} - \overset{5}{\text{CH}_2} - \overset{6}{\text{CH}_2} - \overset{7}{\text{NH}_2}$

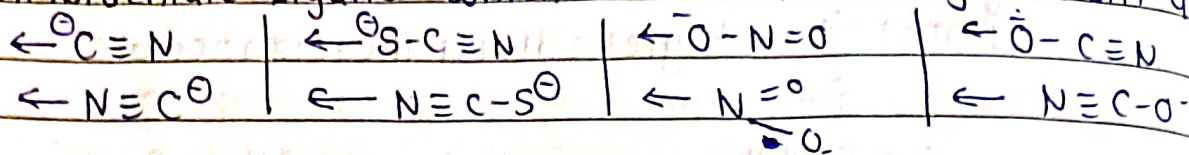


5 member ring

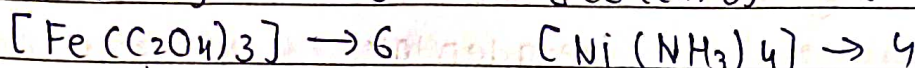
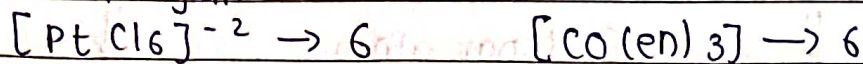


5 member Ring $\rightarrow$ [5]

Ambidentate Ligand. Which can donate through 2 diff atoms

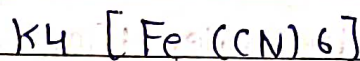


Coordination number - no. of coordinate bond betn central atom & ligand



↓
didentate

Coordination sphere



VS

ionization sphere

ionization sphere

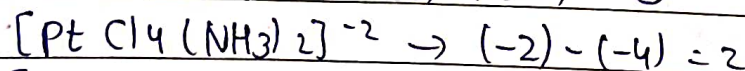
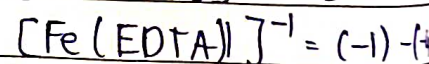
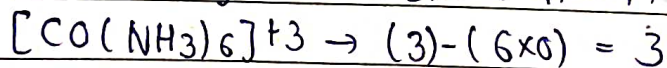
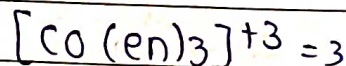
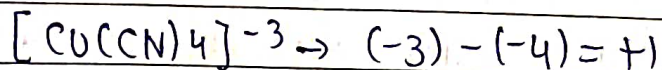
coordination entity

or coordination sphere

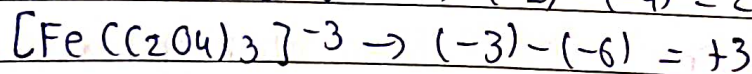
sphere

sphere

Oxidation number - Charge on central atom if all ligands are removed with lone pair

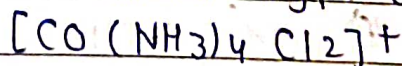


= 3



Homoleptic - only one type of ligand $\rightarrow [\text{Co}(\text{NH}_3)_6]^{+3}$

Heteroleptic - more than one type of ligand



Effective Atomic number (EAN)

no. of e^- on central metal atom forming coordinated bond with ligand

$$\text{EAN} = Z - O.S + 2 \times \text{CN}$$

$$\text{K}_4 [\text{Fe}(\text{CN})_6] \rightarrow 26 - 2 + 2 \times 6 = 36$$

$$\text{K}_3 [\text{Fe}(\text{CN})_6] \rightarrow 26 - 3 + 2 \times 6 = 35$$

Sidgwick Theory - If EAN of metal carbonyl is equal to atomic no. of nearest inert gas then that metal carbonyl is stable

$$[\text{Mn}(\text{CO})_5] \rightarrow 25 - 0 + 2 \times 5 = 35$$

$$[\text{Mn}(\text{CO})_5]^- \rightarrow 25 - (-1) + 2 \times 5 = 36 \rightarrow \text{stable}$$

$$\text{Mn}(\text{CO})_6 \rightarrow \text{EAN} \rightarrow 25 - 0 + 2 \times 6 = 37 \rightarrow e^- \rightarrow 35$$

$$[\text{Mn}(\text{CO})_6]^+ \rightarrow \text{EAN} \rightarrow 25 - 1 + 2 \times 6 = 36 \rightarrow \text{stable}$$

$$\text{Q.) } \text{V}(\text{CO})_6 \rightarrow 23 - 0 + 12 = 35 + e^- \rightarrow 36$$

$$[\text{V}(\text{CO})_6]^- \rightarrow 23 - (-1) + 12 = 36 \rightarrow \text{stable}$$

If $\text{EAN} < \text{atomic no. of inert gas} \rightarrow \text{oxidizing agent}$

$\text{EAN} > \text{atomic no. of inert gas} \rightarrow \text{Reducing agent}$

* Nomenclature

Rules for writing formula.

$\rightarrow$ identify central atom.

$\rightarrow$ write ligands in alphabetical order

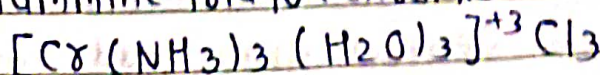
$\rightarrow$ abbreviated ligand $\rightarrow$ consider first letter for alphabetical order

$\rightarrow$ write formula in sq. bracket

$\rightarrow$ no space betn ligand & metal.

$\rightarrow$ If counter ion is not present - charge is indicated on sq. bracket

Triammine triaqua chromium (III) chloride



tris (ethane-1,2-diammine) cobalt (III) Sulphate

Co

Co(en)₃

[Co(en)₃]⁺³ ~~SO₄⁻²~~

[Co(en)₃]₂ (SO₄)₃

Naming of Coordination Compound

→ cation is named first than anion then LCO Rule

→ Ligands are named first (alphabetical order)

then central atom is named & then oxidⁿ no. is written in Roman

Naming of ligand - Cl⁻ → chlorido

anionic ligand C₂O₄⁻² → oxalato

ends in O. O⁻² → oxido

OH⁻ → Hydroxido

Neutral ligand - NH₃ - ammine

H₂O - aqua NO → nitrosyl

CO - carbonyl

→ Prefix - di, tri, tetra → denotes no. of ligand

→ if name of ligand already have (di, tri, tetra) then bis - 2

tris - 3

tetrakis - 4

[NiCl₂(PPh₃)₂]

dichloro (triphenylphosphine) nickel (II)

Complex - cationic → central atom has same name
complex - anionic → central atom add rate 1 in end

$[\text{Co}(\text{SCN})_4]^{-2} \rightarrow \text{Cobaltate}$

$[\text{Fe}(\text{CN})_6]^{-4} \rightarrow \text{Ferriate}$

$[\text{Pt}(\text{NH}_3)_6]^{+2} \rightarrow \text{Platinum}$

$[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]^{+3} \rightarrow \text{triamminetriaquachromium (III) chloride}$

$[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3 \rightarrow$

$[\text{Co}(\text{en})_3]^{+3} \text{SO}_4^{-2} \rightarrow \text{tris (ethylenediamine) cobalt sulphate}$
(ethane 1,2, diammine)

$[\text{Ag}(\text{NH}_3)_2]^+ [\text{Ag}(\text{CN})_2]^-$

diammine Silver (I) dicyanoargentate (I)

* Isomerism

* Structural isomerism

i) Ionization

$\rightarrow$ comp. which give
diff. ions

$[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$

$[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

ii) Solvate

$\rightarrow$ no. of water molecules inside
4 outside coordination sphere
is different

$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

$[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$

iii) Linkage

$\rightarrow$ ambidentate ligand

$[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

$[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$

iv) Coordination

$\rightarrow$ cation & anion both are
complex

$[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$

$[\text{Co}(\text{CN})_6] [\text{Cr}(\text{NH}_3)_6]$

* Stereo isomer

i) Geometrical isomer

Tetrahedral $\times$

Sq. planar $\checkmark$

Octahedral $\checkmark$

Coordination no. = 4

$MA_4 = 0$

$MA_3B = 0$

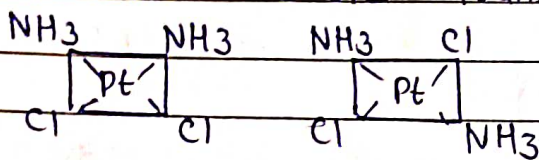
$MA_2B_2 = 2 \begin{cases} \text{cis} \\ \text{trans} \end{cases}$

$MA_2BC = 2 \begin{cases} \text{cis} \\ \text{trans} \end{cases}$

$MABCD = 3 \begin{cases} \text{cis} \\ \text{trans} \end{cases}$

$[Pt(NH_3)_2Cl_2]$

$MA_2B_2 \rightarrow 2 GI \begin{cases} \text{cis} \\ \text{trans} \end{cases}$

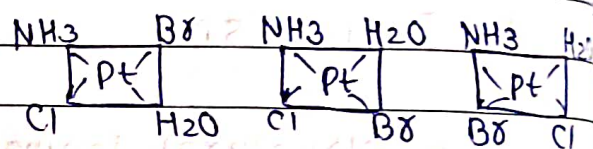


cis

trans

$[Pt(NH_3)(H_2O)(Cl)(Br)]$

$MABCD \rightarrow 3 GI \begin{cases} \text{cis} \\ \text{trans} \end{cases}$



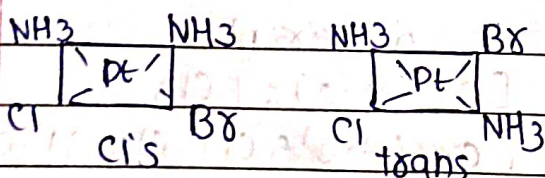
(ab cd)

(ac bd)

(ad bc)

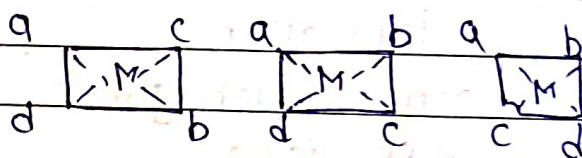
$Pt(NH_3)_2ClBr$

$MA_2BC = 2 GI \begin{cases} \text{cis} \\ \text{trans} \end{cases}$



cis

trans



trans

cis

cis

Bidentate

$[Pt(en)_2]^{+2}$

$[Pt(gly)_2]$

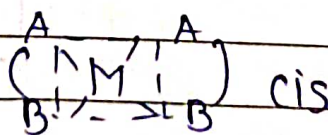
$[Pt(gly)(NH_3)Cl]$

$[M(AA)_2]^{+2}$

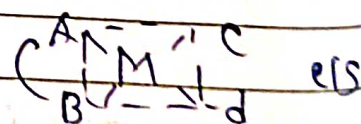
$[M(AB)_2]$

$M(AB)cd$

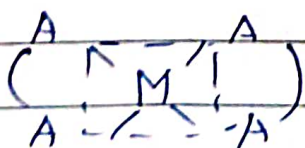
2 donor atom of bidentate ligand never lie at 180°



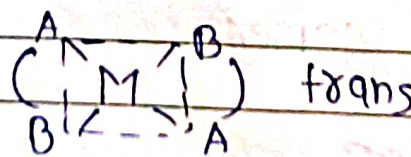
cis



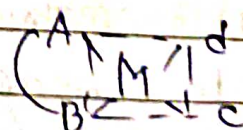
cis



No GI



trans



trans

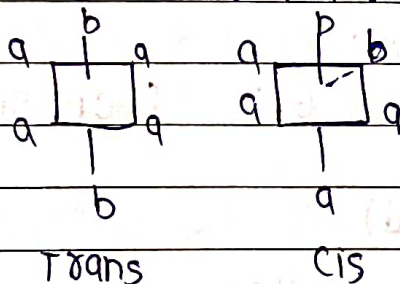
Geometrical & Optical isomer in Octahedral

	GI	Pair of enantiomer	no. of stereoisomer (GI + OE)
MA ₆	0		
MA ₅ b	0		
MA ₄ b ₂	2		
MA ₄ bc	2		
MA ₃ b ₃	2		
MA ₃ b ₂ c	3		
MA ₃ bcd	4	+	5
MA ₂ b ₂ c ₂	5	+	6
MA ₂ bcd ₂	6	+	8
Mabcde ₂	9	+	15
Mabcdef	15	+	30

[Pt(NH₃)₄Cl₂] or [Co(NH₃)₄Cl₂]⁺

MA₄b₂

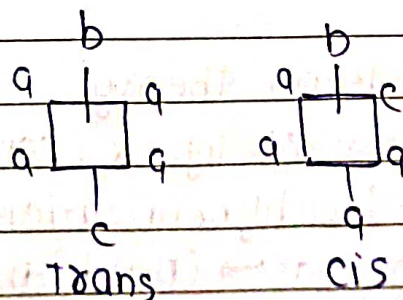
GI = 2 $\begin{cases} \text{cis} \\ \text{trans} \end{cases}$



[Pt(NH₃)₄ClBr]

MA₄bc

GI = 2 $\begin{cases} \text{cis} \\ \text{trans} \end{cases}$

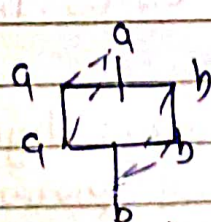


MA₃b₃

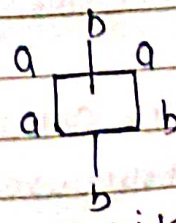
[Co(NH₃)₃Cl₃]

[Co(NH₃)₃(NO₂)₃]

2 GI $\begin{cases} \text{fac} \\ \text{mer} \end{cases}$

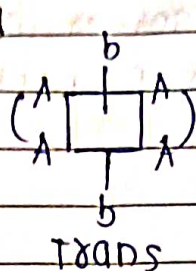
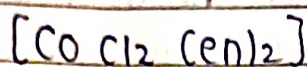


facial isomer (fac)

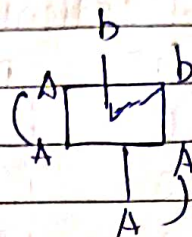


meridional isomer (mer)

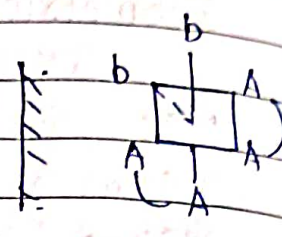
Bidentate ligand



TRANS



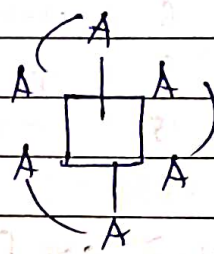
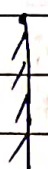
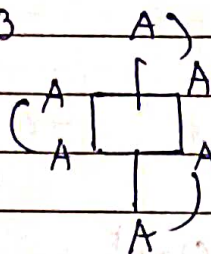
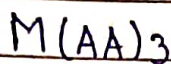
cis



(chiral molecule)

8 shows O.I

non superimposable mirror images



Chixal

Shows G.I

Non superimposable mirror images

(d)

(2)

dextrorotatory

PPI $\downarrow$ (↓ve)

rotation

levorotatory

PPI S (-ve)

rotation

* Valence - Bond Theory:

Under influence of ligand, central atom uses two types of orbital for hybridization.

1) inner orbital $\rightarrow (n-1)d, ns, np$ } vacant orbital
2) outer orbital $\rightarrow ns, np, nd$ }

2) Outer orbitals $\rightarrow ns, np, nd$

vacant orbital undergo hybridization & gives octahedral & sq. planar geometry

CN=4 (sp^3 - Tetrahedral)

dsp^2 - Sq. planar

CN=6 (d^2sp^3 - Octahedral)

sp^3d^2 - Octahedral

SFL $\rightarrow$ Pairing occurs

WFL $\rightarrow$ Pairing do not occurs

Spectrochemical Series

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA < NH_3 < en < NO_2^- < CN^- < CO$

Carbon / nitrogen donor - SFL

Oxygen / halogen donor - WFL

Imp point

$\rightarrow$ All ligand act as SFL with 4d/5d metal ions

$\rightarrow F^-$ act as SFL with Ni^{+4}

$\rightarrow H_2O$ & $C_2O_4^{2-}$ act as SFL Co^{+3}

$\rightarrow NH_3 \rightarrow WFL \rightarrow Fe^{+2}$ & Mn^{+2}

$\rightarrow H_2O \rightarrow SFL \rightarrow Cu^{+2}$ (CN=4)

$[Co(NH_3)_6]^{+3}$

$Co^{+3} \rightarrow 3d^6$

$Co-3d^1 4s^2$

$\downarrow$

CN=6

SFL

Pairing $\checkmark$

$1\uparrow\downarrow 1\uparrow\downarrow 1\uparrow\downarrow \square \square \square \square$

(n-1) d

ns

np

d^2sp^3

inner orbital complex

low spin / spin paired

diamagnetic

$[CoF_6]^{-3} \rightarrow Co^{+3} \rightarrow 3d^6$

$\hookrightarrow WFL$

Pairing $\times$

$1\uparrow\downarrow 1\uparrow\downarrow 1\uparrow\downarrow 1\uparrow\downarrow 1\uparrow\downarrow 1\uparrow\downarrow$

ns

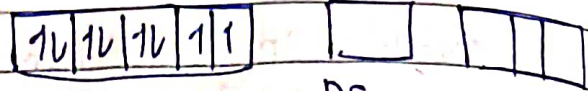
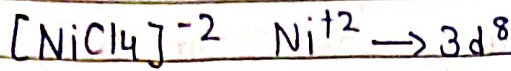
np

nd

paramagnetic

sp^3d^2

Outer orbital complex / high spin / spin free complex



↳ WFL

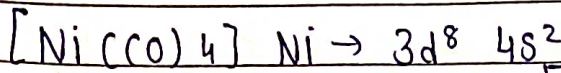
ds

np

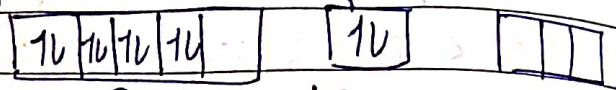
Pairing x

sp³ tetrahedral

paramagnetic



↳ SFL



Pairing ✓

3d

4s

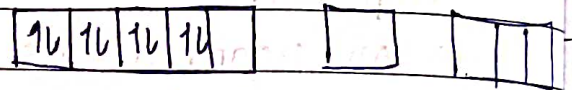
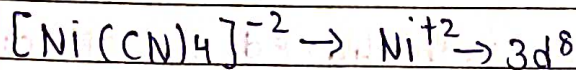
4p



sp³

tetrahedral

diamagnetic



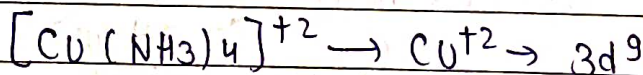
SFL

dsp²

Pairing ✓

Sq. planar

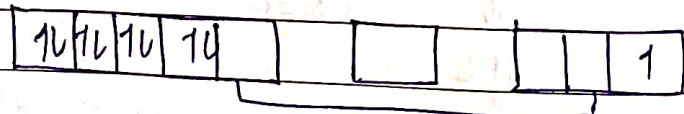
diamagnetic



SFL



d e⁻ excites



dsp²

Sq. planar

paramagnetic

Magnetic property of C.C

Octahedral Case

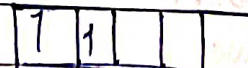


SFL/WFL

↓

always d² sp³

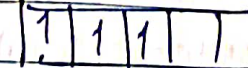
inner orbital



no pairing

Complex

is needed



$d^8 d^9 d^{10}$

1↓	1↓	1↓	↑	↑
----	----	----	---	---

SFL/WFL, always $sp^3 d^2$



Outer orbital

1↓	1↓	1↓	1↓	↑
----	----	----	----	---

No pairing

Complex

occurs

1↓	1↓	1↓	1↓	↑↓
----	----	----	----	----

Magnetic moment

$[Mn(CN)_6]^{-3} \rightarrow Mn^{+3} \rightarrow 3d^4$

Pairing

↑	↑	↑	↑
---	---	---	---

1↓	1↓	↑	↑
----	----	---	---

Unpaired $e^- = 2$

$$MM = \sqrt{n(n+2)} \text{ BM}$$

$$= \sqrt{2 \times 4}$$

$$= \sqrt{8} \text{ BM}$$

$[MnCl_6]^{-3} \rightarrow Mn^{+3} \rightarrow 3d^4$

para. $\rightarrow$ WFL - no pairing

Unpaired $e^- = 4$

$$MM = \sqrt{4(4+2)} \text{ BM}$$

$$= \sqrt{24} \text{ BM}$$

$[Fe(CN)_6]^{-3} \rightarrow Fe^{+3} \rightarrow 3d^5$

para $\rightarrow$ Pairing

↑	↑	↑	↑	↑
---	---	---	---	---

1↓	1↓	↑	↑
----	----	---	---

$$(n=1) \quad MM = \sqrt{3} \text{ BM}$$

* CFT

$\rightarrow$ electrostatic Model

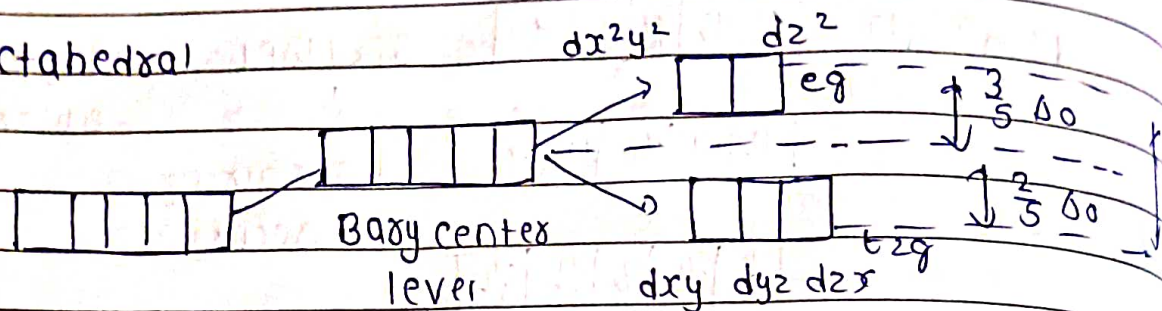
$\rightarrow$ Central metal $\rightarrow$ +ve charge } electrostatic

$\rightarrow$ Ligand $\rightarrow$ -ve charge

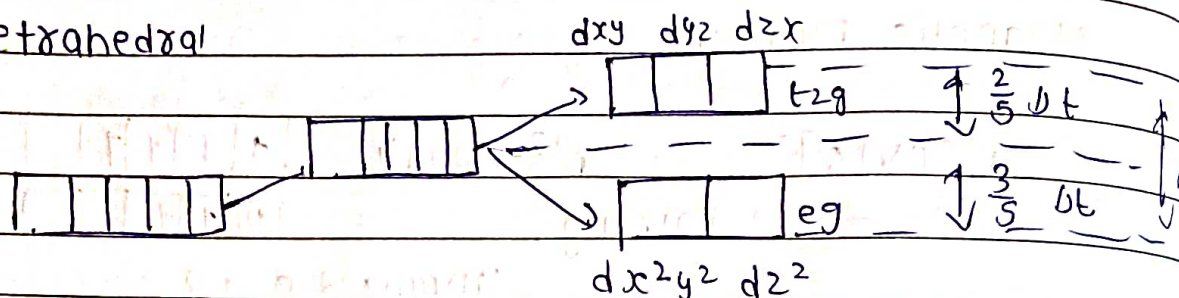
$\rightarrow$ Degeneracy of d orbital breaks when ligand approaches central atom
repulsion betⁿ e^- of d orbital & lone pair of ligand

So d orbital splitting occurs

Octahedral



Tetrahedral



$$\Delta t = \frac{4}{9} \Delta o$$

$\Delta o < P \rightarrow$ WFL $\rightarrow$ pairing not occur

$\Delta o > P \rightarrow$ SFL $\rightarrow$ pairing occur

Factors affecting CFSE

$\Delta o \propto$ charge on CMA

$\Delta o \propto z_{eff}$ of CA

$3d < 4d < 5d \rightarrow$ CFSE

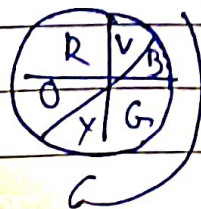
$\Delta o \propto$ Strength of ligand

Octahedral $>$ tetrahedral ($\Delta t = \frac{4}{9} \Delta o$)

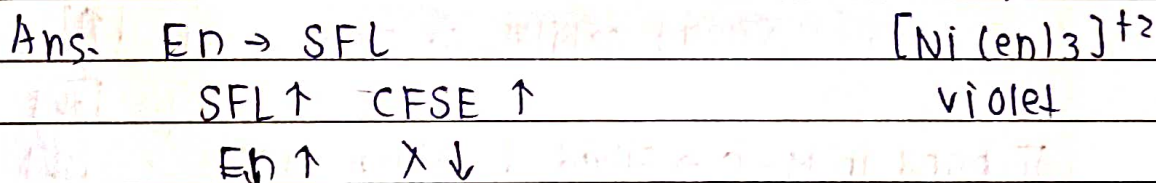
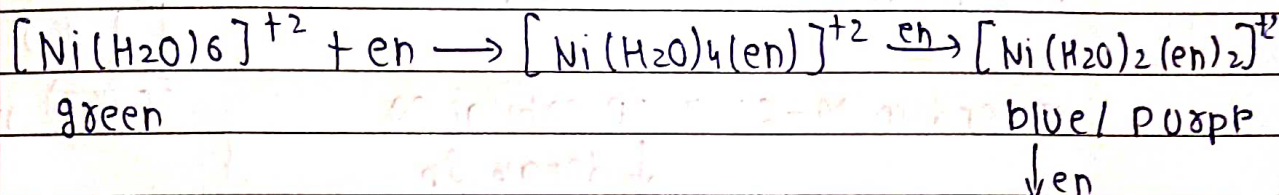
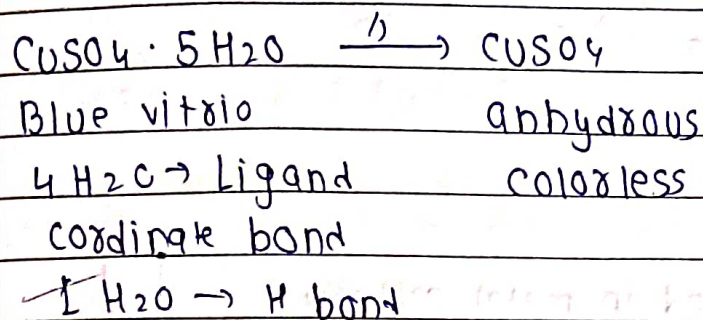
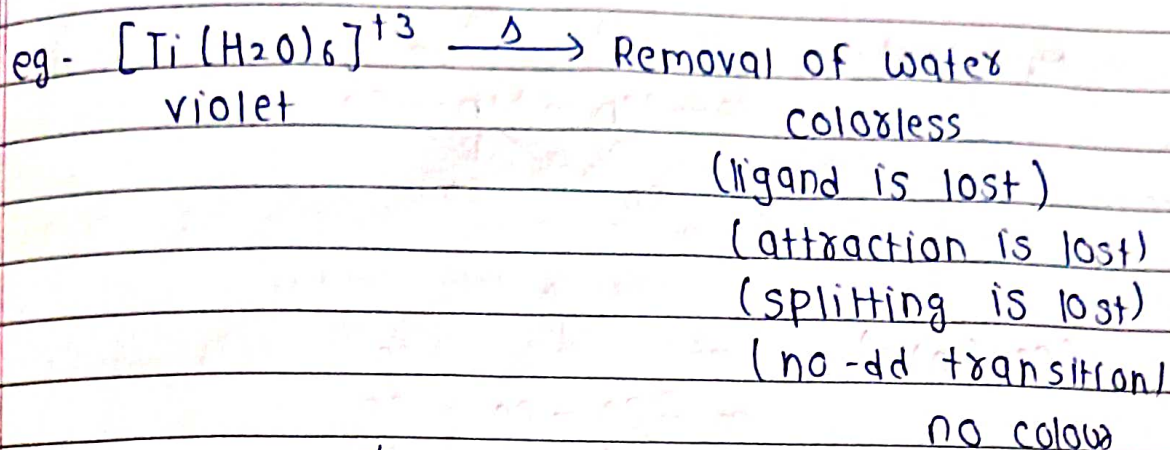
Color

e^- jump $(t_{2g} - e_g)$ } d-d transition
 $(e_g - t_{2g})$ } color

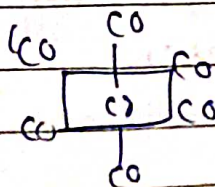
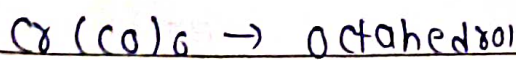
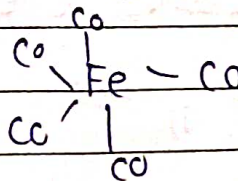
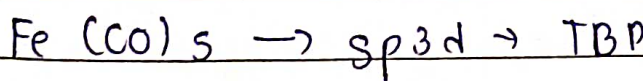
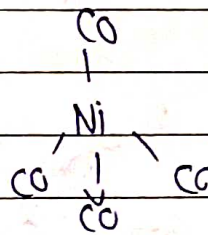
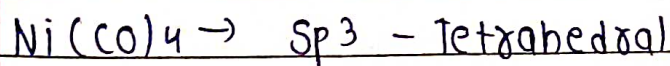
$\rightarrow$ It always shows complimentary colour

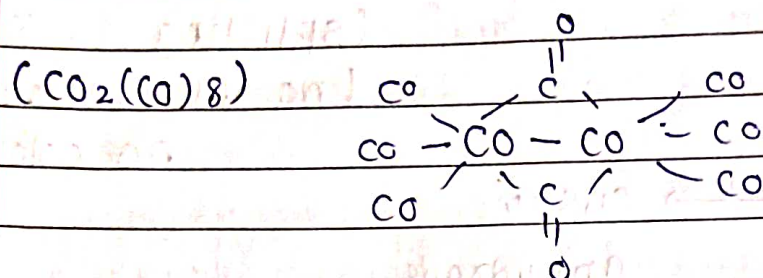
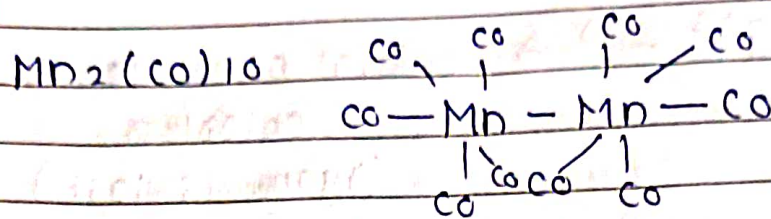


$\lambda \uparrow$



Bonding in metal carbonyl:





Synergic bond.

metal-carbon bond in metal carbonyl has both σ and π bond.

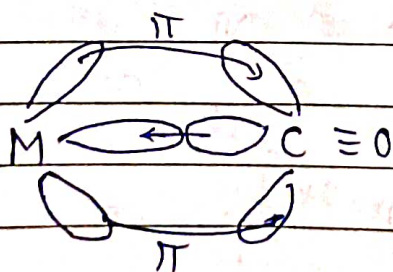
σ bond in $M-C$: π 2p orbital CO
 $\downarrow$ donate lp

vacant orbital of metal

π bond in $M-C \rightarrow$ Filled d orbital of M

$\downarrow$ donate

π^* (antibonding) molecular orbital of CO



Synergic bond
 Strong

